DSA 8420 Spring 2025
Homework 7

Due March 4, 2025

1. (20 points) Follow the instructions in the slides on Canvas to install Jupyter Notebook, Pyomo, and GLPK on your computer. Download the Jupyter Notebook file from Canvas under Video 3.1 and run the code. Submit a screenshot confirming that the linear program has been successfully solved on your computer.

2. Consider the following brewery problem, which was discussed in Video 2.1.

   Microbrewers Incorporated makes four products called Light, Dark, Ale, and Premium. These products are made using the resources of water, malt, hops, and yeast. Microbrewers has a free supply of water, so it is the amount of the other resources that restricts production capacity. The technology table gives the pounds of each resource required in the production of 1 gal of each product. The problem faced by Microbrewers Inc. is to decide how much of each product it should make in order to maximize its revenue.

	Light	Dark	Ale	Premium	Available
Malt	1	1	0	3	50 lb
Hops	2	1	2	1	150 lb
Yeast	1	1	1	4	80 lb
Revenue	\$6	\$5	\$3	\$7	

Table 1: Technology Table for Microbrewers, Inc.

To solve the problem, we define the decision variables as

$$
\begin{aligned}
x_1 &= \text{Light (gallons)} \\
x_2 &= \text{Dark (gallons)} \\
x_3 &= \text{Ale (gallons)} \\
x_4 &= \text{Premium (gallons)}
\end{aligned}
$$

The linear programming model is as follows.

$$
\begin{array}{lrcl}
\max & 6x_1 + 5x_2 + 3x_3 + 7x_4 & & \text{profit(\$)} \\
\text{s.t.} & x_1 + x_2 \phantom{{}+2x_3} + 3x_4 \leq 50 & & \text{malt (pounds)} \\
 & 2x_1 + x_2 + 2x_3 + x_4 \leq 150 & & \text{hops (pounds)} \\
 & x_1 + x_2 + x_3 + 4x_4 \leq 80 & & \text{yeast (pounds)} \\
 & x_1,\, x_2,\, x_3,\, x_4 \geq 0 & &
\end{array}
$$

In the provided Jupyter Notebook (brewery.ipynb), the model is implemented in two formats: a concrete model and an abstract model. Compare the two models and complete the missing code.

(a) (15 points) In the concrete model, fill in the code which builds the objective function. Run the code. Submit a screenshot of your completed code **to Canvas** and a screenshot of the final output **to Gradescope**.

(b) (15 points) In the abstract model, fill in the code which builds the yeast constraint and the code which prints the optimal solution. Run the code. Submit a screenshot of your completed code **to Canvas** and a screenshot of the final output **to Gradescope**.